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Date: 1/15/24

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Course: Calculus Beta

Assignment: MyLab Math - Uge 1

Find the derivative of the function.

$$y = x^5 e^x$$

Use the product rule, $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$, to take the derivative.

Let $x^5 = u$ and let $e^x = v$. Take the derivative of e^x and find $u \frac{dv}{dx}$.

$$\begin{aligned} u \frac{dv}{dx} &= x^5 \frac{d}{dx} e^x \\ &= x^5 e^x \end{aligned}$$

Now take the derivative of x^5 and find $v \frac{du}{dx}$.

$$\begin{aligned} v \frac{du}{dx} &= e^x \frac{d}{dx} x^5 \\ &= 5e^x x^4 \end{aligned}$$

Add the two terms $u \frac{dv}{dx}$ and $v \frac{du}{dx}$ together to find the derivative.

$$\begin{aligned} \frac{d}{dx}(x^5 e^x) &= u \frac{dv}{dx} + v \frac{du}{dx} \\ &= x^5 e^x + 5x^4 e^x \end{aligned}$$